

Qualification Pack



Mine Welder

Electives: Underground Metal/ Opencast/ Underground Coal

Options: Underground Gassy Mines

QP Code: MIN/Q3201

Version: 3.0

NSQF Level: 4

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MIN/Q3201: Mine Welder

Brief Job Description

A mine welder is responsible for joining various types of metallic frames, structures, jigs, plates, sheets etc. using heating and melting process created through electrical power and gaseous discharge, maintaining process parameters, conducting quality checks on output product and maintaining a safe and healthy working environment.

Personal Attributes

This job requires quality consciousness, safety orientation, physique to sustain strenuous conditions, ability to use fingers, hands and feet with ease to complete the assigned task (Dexterity).

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [MIN/N3201: Prepare the welding machine, auxiliaries and work pieces for the welding process](#)
2. [MIN/N3202: Conduct the welding process and weld the work piece](#)
3. [MIN/N3203: Post welding operations activities](#)
4. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Electives(mandatory to select at least one):

Elective 1: Underground Metal

This unit is about adhering to health, safety and environmental guidelines in Underground Metalliferous Mines during execution of various tasks, operations and maintenance.

1. [MIN/N1702: Follow Health, Safety and Environmental guidelines for Underground Metalliferous Mines](#)

Elective 2: Opencast

This unit is about adhering to health, safety and environmental guidelines in the Open Cast Mines during execution of various tasks, operations and maintenance.

1. [MIN/N1703: Follow Health, Safety, and Environmental Guidelines for opencast mines](#)

Elective 3: Underground Coal

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This unit is about adhering to health, safety and environmental guidelines in Underground Coal Mines (UCM) during execution of various tasks, operations and maintenance.

1. [MIN/N1704: Follow Health, Safety, and Environmental guidelines for underground coal mines](#)

Options(Not mandatory):

Option : Underground Gassy Mines

This OS unit is about Special preparation for Welding at U/G Gassy Mines.

1. [MIN/N3204: Special preparation for Welding at U/G Mines](#)

Qualification Pack (QP) Parameters

Sector	Mining
Sub-Sector	Engineering Services
Occupation	Mechanical Services
Country	India
NSQF Level	4
Credits	21
Aligned to NCO/ISCO/ISIC Code	NCO-2015/7212.0200
Minimum Educational Qualification & Experience	12th grade Pass (or equivalent) OR 11th grade pass (with 1.5 years relevant experience in Mining Industry) OR 10th grade pass (with 3 years relevant experience in Mining Industry) OR Previous relevant Qualification of NSQF Level (3:Mine Mechanic/Fitter with 3 years relevant experience in Mining Industry)
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	Not Applicable
Minimum Job Entry Age	20 Years



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Last Reviewed On	NA
Next Review Date	06/05/2028
NSQC Approval Date	08/05/2025
Version	3.0
Reference code on NQR	QG-04-MG-04062-2025-V2-SCMS
NQR Version	1.0

Remarks:

Total Notional Duration: 480 Hours; Additional 30 hours for Optional NOS= Domain Theory Duration: 100 Hours; Additional 10 hours for Optional NOS Theory Duration+ Domain Practical Duration: 350 Hours; Additional 20 hours for Optional NOS Practical Duration+ Employability Skills Duration: 30 Hours

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MIN/N3201: Prepare the welding machine, auxiliaries and work pieces for the welding process

Description

This OS unit is about recognizing the work requirement and preparing the welding machine, auxiliary apparatus like transformers, gas cylinder, flux wires etc. and metal work pieces (jigs) for the welding process.

Scope

The scope covers the following :

- Recognizing the welding requirements, welding equipment and parameters
- Arrange the material and equipment
- Clean the welding equipment and setup the equipment
- Prepare the surface of the part (work pieces)

Elements and Performance Criteria

Recognizing the welding requirements, welding equipment and parameters

To be competent, the user/individual on the job must be able to:

- PC1.** recognize the right welding methodology and process to be adopted for completing the work order
- PC2.** refer all process manuals/ work instructions/Standard Operating Procedures (SOP) to understand the right welding methodology and process
- PC3.** discuss with the supervisor for given task/work
- PC4.** identify the various welding parameters like temperature, pressure, electrode type, electrode distance, process cycle time etc. before starting the welding process, as mentioned in the work Instructions/ SOP manual
- PC5.** identify the material required and the equipment availability for executing the activity
- PC6.** compare various type of electrode in terms of electrode material and thickness, filler material and flux which will be required for the selected welding process before the initiation of the welding process

Arrange the material and equipment

To be competent, the user/individual on the job must be able to:

- PC7.** arrange the correct type of electrodes as per material wise, dimension wise, filler material wise etc. used for the welding process
- PC8.** arrange the material and the equipment required for executing the activity
- PC9.** procure all required material from the store before starting the welding process

Clean the welding equipment and setup the equipment

To be competent, the user/individual on the job must be able to:

- PC10.** clean the surface of the electrodes and remove dust and any other impurities from the welding gun

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- PC11.** clean the other welding machine auxiliaries(welding transformer, gas discharge unit, flux wire) before the initiation of the welding process, as mentioned in the work instructions/ Standard Operating Procedures(SOP)
- PC12.** setup the welding apparatus as per the selected welding process and the internal SOPs/ Work instructions and the setting standard for the machine

Prepare the surface of the part (work pieces)

To be competent, the user/individual on the job must be able to:

- PC13.** clean the surface of the metal parts (work pieces) which need to be joint
- PC14.** prepare the edge for the strongest possible weld using techniques like machining, chipping, grinding, oxy- acetylene cutting and carbon arc cutting
- PC15.** ensure that the parameters for edge parameters are as per the desired specifications like speed, cost, adaptability etc.
- PC16.** take the work permission/authorization from mines manager if required

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** job-specific documents e.g. daily maintenance checklist and importance of the same
- KU2.** risk and impact of not following defined procedures/work instructions
- KU3.** the hierarchy for reporting identified problems
- KU4.** cost of equipment and loss for the company that results from damage of equipment
- KU5.** implications of delays in the process
- KU6.** handover and takeover procedures of the mine welder according to company's SOP
- KU7.** safety guidelines specified by Directorate General of Mines Safety (DGMS) specific to welding works
- KU8.** different types of mines and detail of the mine one is working in
- KU9.** benching in quarries, dressing of overhangs, undercuts, fencing
- KU10.** importance of first aid and hygiene
- KU11.** code of practice in specific areas of the mine
- KU12.** standing orders in force at the mine
- KU13.** importance of safety in the vicinity of machinery
- KU14.** about shot-firing / blasting related safety regulations including taking shelter during blasting
- KU15.** duties of workmen under the Mines Act-1952
- KU16.** provision of compensation and working hours, leaves, etc. as per Mines Act-1952
- KU17.** the outcome of violation of safety procedures
- KU18.** emergency response /disaster management plan prepared by the organization
- KU19.** different types of welding processes and associated equipment: GMAW, FCAW, SMAW, GTAW, TIG, etc.
- KU20.** different cleaning methods for electrodes, metal surfaces etc.
- KU21.** how to use measuring instruments like vernier calipers, micrometers
- KU22.** different types of welding joints

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- KU23.** procedure to issue and receive material from the stock
- KU24.** the impact of various physical parameters like temperature, pressure, electrode distance, electric current, voltage on the properties of final output product like durability, ductility, surface finish etc.
- KU25.** basic principles of geometric and drawing
- KU26.** methods of edge preparation and associated equipment

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** note down observations (if any)
- GS2.** read and interpret symbols and readings
- GS3.** read and interpret sketches and engineering drawings
- GS4.** read information documents
- GS5.** discuss task lists, schedules and activities
- GS6.** effectively communicate, listen and comprehend the information given by various sources about the site
- GS7.** make decisions pertaining to the concerned area of work
- GS8.** plan and organize the work order and tasks
- GS9.** organize all operation and service manuals so that sorting/ accessing information is easy
- GS10.** detect problems in day to day tasks
- GS11.** discuss possible solution with the supervisor for problem solving
- GS12.** make decisions in emergency conditions
- GS13.** follow instructions and work on areas of improvement identified
- GS14.** complete the assigned tasks timely
- GS15.** use reasoning skills to identify and resolve basic problems
- GS16.** analyze and detect any potential problems which could arise during operation
- GS17.** respect persons with disabilities
- GS18.** explain the importance of gender-sensitization at work site
- GS19.** state basic laws, acts and provisions defined for Persons with Disability (PwD) by the statutory bodies

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Recognizing the welding requirements, welding equipment and parameters</i>	12	19	-	7
PC1. recognize the right welding methodology and process to be adopted for completing the work order	2	3	-	1
PC2. refer all process manuals/ work instructions/Standard Operating Procedures (SOP) to understand the right welding methodology and process	2	3	-	1
PC3. discuss with the supervisor for given task/work	2	3	-	1
PC4. identify the various welding parameters like temperature, pressure, electrode type, electrode distance, process cycle time etc. before starting the welding process, as mentioned in the work Instructions/ SOP manual	2	3	-	1
PC5. identify the material required and the equipment availability for executing the activity	2	4	-	2
PC6. compare various type of electrode in terms of electrode material and thickness, filler material and flux which will be required for the selected welding process before the initiation of the welding process	2	3	-	1
<i>Arrange the material and equipment</i>	5	9	-	3
PC7. arrange the correct type of electrodes as per material wise, dimension wise, filler material wise etc. used for the welding process	2	3	-	1
PC8. arrange the material and the equipment required for executing the activity	1	3	-	1
PC9. procure all required material from the store before starting the welding process	2	3	-	1
<i>Clean the welding equipment and setup the equipment</i>	5	9	-	4
PC10. clean the surface of the electrodes and remove dust and any other impurities from the welding gun	1	3	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. clean the other welding machine auxiliaries(welding transformer, gas discharge unit, flux wire) before the initiation of the welding process, as mentioned in the work instructions/ Standard Operating Procedures(SOP)	2	3	-	2
PC12. setup the welding apparatus as per the selected welding process and the internal SOPs/ Work instructions and the setting standard for the machine	2	3	-	1
<i>Prepare the surface of the part (work pieces)</i>	8	13	-	6
PC13. clean the surface of the metal parts (work pieces) which need to be joint	2	3	-	1
PC14. prepare the edge for the strongest possible weld using techniques like machining, chipping, grinding, oxy- acetylene cutting and carbon arc cutting	2	4	-	2
PC15. ensure that the parameters for edge parameters are as per the desired specifications like speed, cost, adaptability etc.	2	3	-	2
PC16. take the work permission/authorization from mines manager if required	2	3	-	1
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	MIN/N3201
NOS Name	Prepare the welding machine, auxiliaries and work pieces for the welding process
Sector	Mining
Sub-Sector	Engineering Services
Occupation	Mechanical Services
NSQF Level	4
Credits	5
Version	2.0
Last Reviewed Date	08/05/2025
Next Review Date	06/05/2028
NSQC Clearance Date	08/05/2025

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MIN/N3202: Conduct the welding process and weld the work piece

Description

This OS is about conducting welding operation as per the methodology selected for welding and the Standard Operating Procedures (SOP) defined by the organization and the outcome of the work order.

Scope

The scope covers the following :

- Install the welding work pieces on the welding apparatus
- Check the operations of the welding machines and auxiliaries and conduct a test process
- Perform the actual welding process
- Monitor process parameters to ensure error free welding process
- Measure both welded pieces and remove welding inconsistency

Elements and Performance Criteria

Install the welding work pieces on the welding apparatus

To be competent, the user/individual on the job must be able to:

- PC1.** hold the parts (Jigs) which need to be welded together using a clamp and align them with the electrodes as per the job requirement
- PC2.** install the work pieces on the welding apparatus keeping in mind the electrodes distance, contact area, pressure, temperature application etc as specified in the welding SOP/ Control plan documents/work Instructions and instructed by the supervisor

Check the operations of the welding machines and auxiliaries and conduct a test process

To be competent, the user/individual on the job must be able to:

- PC3.** check for operation of core welding equipment like welding gun, welding transformer, gas cylinders and gas discharge as per setup documentation
- PC4.** identify shift direction hazard
- PC5.** conduct destructive and non- destructive test activity to ensure conformance to the SOPs/ work instructions
- PC6.** inform supervisor to make modifications in the welding parameters as per the test activity outcomes and the prescribed standard for destructive/ non destructive Tests

Perform the actual welding process

To be competent, the user/individual on the job must be able to:

- PC7.** adjust the current/ voltage, temperature application as per the welding requirement and the activity test conducted earlier so that the desired heat can be created for the welding process
- PC8.** check for the positioning of the spot and the welding gun as per the work instructions and the work order
- PC9.** / hold the filler metal/ Flux material wire and the Welding Gun at the recommended angle and distance mentioned in the setup document, keeping the work pieces stationary to ensure the required melting of base metal

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PC10. ensure the flow of filler material/ gas discharge as per the welding standard prescribed in the SOP/ Work Instructions

Monitor process parameters to ensure error free welding process

To be competent, the user/individual on the job must be able to:

PC11. monitor the welding process (Pressure, Temperature, gas discharge flow, electrode force, electrode distance etc) by observing the readings on the panels/ measuring instruments to prevent any harm to the work pieces due to overheating, burning, over melting, change in applied pressure etc

PC12. note down the observations in the prescribed format

Measure both welded pieces and remove welding inconsistency

To be competent, the user/individual on the job must be able to:

PC13. measure the final welded piece and compare the dimensions as prescribed in the work order engineering drawing

PC14. ensure that the assistant operators/ helpers remove extra material by using chippers, grinders etc., in case the parts are not as per the given measurements

PC15. ensure hammering of the bulges to give the work pieces the desired shape, in case of any dents or bulges

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. job-specific documents e.g. daily maintenance checklist and importance of the same

KU2. risk and impact of not following defined procedures/work instructions

KU3. the hierarchy for reporting identified problems

KU4. cost of equipment and loss for the company that results from damage of equipment

KU5. implications of delays in the process

KU6. handover and takeover procedures of the welding work according to company's SOP

KU7. safety guidelines specified by Directorate General of Mines Safety (DGMS) specific to welding operations

KU8. different types of mines and detail of the mine one is working i

KU9. benching in quarries, dressing of overhangs, undercuts, fencing

KU10. importance of first aid and hygiene

KU11. code of practice in specific areas of the mine

KU12. standing orders in force at the mine

KU13. importance of safety in the vicinity of machinery

KU14. about shot-firing / blasting related safety regulations including taking shelter during blasting

KU15. duties of workmen under the Mines Act-1952

KU16. provision of compensation and working hours, leaves, etc. as per Mines Act-1952

KU17. the outcome of violation of safety procedures

KU18. emergency response /disaster management plan prepared by the organization

KU19. different types of welding processes and associated equipment

KU20. different types of joints used in welding

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- KU21.** different cleaning methods for electrodes, metal surfaces etc
- KU22.** the methods of using instruments like Vernier calipers, Micrometers, rulers and other inspection tools
- KU23.** various national and International welding standard used in mining sector in India
- KU24.** how to visually represent the final product outcome
- KU25.** different types of defects in welding and their impact
- KU26.** potential health and safety hazards and related Safety precautions to be undertaken during the welding process
- KU27.** basic chemical properties of material used for electrodes, flux, welding gases etc.
- KU28.** basic electrical laws and working of welding transformers, capacitors etc..
- KU29.** DC welding machines, checking voltage of machines etc
- KU30.** welding gauges, different type of gauges etc.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** note down observations (if any)
- GS2.** read and interpret symbols and readings
- GS3.** read information documents
- GS4.** read and interpret sketches and engineering drawings
- GS5.** represent the final product output and hence decide on the key steps to be followed for welding
- GS6.** discuss task lists, schedules and activities
- GS7.** effectively communicate, listen and comprehend the information given by various sources about the site
- GS8.** make decisions pertaining to the concerned area of work
- GS9.** plan and organize the work order and tasks
- GS10.** organize all operation and service manuals so that sorting/ accessing information is easy
- GS11.** detect problems in day to day tasks
- GS12.** discuss possible solution with the supervisor for problem solving
- GS13.** make decisions in emergency conditions
- GS14.** follow instructions and work on areas of improvement identified
- GS15.** complete the assigned tasks timely
- GS16.** use reasoning skills to identify and resolve basic problems
- GS17.** analyze and detect any potential problems which could arise during operation
- GS18.** respect persons with disabilities
- GS19.** explain the importance of gender-sensitization at work site
- GS20.** state basic laws, acts and provisions defined for Persons with Disability (PwD) by the statutory bodies

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Install the welding work pieces on the welding apparatus</i>	4	8	-	4
PC1. hold the parts (jigs) which need to be welded together using a clamp and align them with the electrodes as per the job requirement	2	4	-	2
PC2. install the work pieces on the welding apparatus keeping in mind the electrodes distance, contact area, pressure, temperature application etc as specified in the welding SOP/ Control plan documents/work Instructions and instructed by the supervisor	2	4	-	2
<i>Check the operations of the welding machines and auxiliaries and conduct a test process</i>	8	13	-	5
PC3. check for operation of core welding equipment like welding gun, welding transformer, gas cylinders and gas discharge as per setup documentation	2	3	-	1
PC4. identify shift direction hazard	2	3	-	1
PC5. conduct destructive and non- destructive test activity to ensure conformance to the SOPs/ work instructions	2	4	-	2
PC6. inform supervisor to make modifications in the welding parameters as per the test activity outcomes and the prescribed standard for destructive/ non destructive Tests	2	3	-	1
<i>Perform the actual welding process</i>	8	13	-	5
PC7. adjust the current/ voltage, temperature application as per the welding requirement and the activity test conducted earlier so that the desired heat can be created for the welding process	2	3	-	1
PC8. check for the positioning of the spot and the welding gun as per the work instructions and the work order	2	4	-	2

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC9. / hold the filler metal/ Flux material wire and the Welding Gun at the recommended angle and distance mentioned in the setup document, keeping the work pieces stationary to ensure the required melting of base metal	2	3	-	1
PC10. ensure the flow of filler material/ gas discharge as per the welding standard prescribed in the SOP/ Work Instructions	2	3	-	1
<i>Monitor process parameters to ensure error free welding process</i>	4	7	-	3
PC11. monitor the welding process (Pressure, Temperature, gas discharge flow, electrode force, electrode distance etc) by observing the readings on the panels/ measuring instruments to prevent any harm to the work pieces due to overheating, burning, over melting, change in applied pressure etc	2	4	-	2
PC12. note down the observations in the prescribed format	2	3	-	1
<i>Measure both welded pieces and remove welding inconsistency</i>	6	9	-	3
PC13. measure the final welded piece and compare the dimensions as prescribed in the work order engineering drawing	2	3	-	1
PC14. ensure that the assistant operators/ helpers remove extra material by using chippers, grinders etc., in case the parts are not as per the given measurements	2	3	-	1
PC15. ensure hammering of the bulges to give the work pieces the desired shape, in case of any dents or bulges	2	3	-	1
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	MIN/N3202
NOS Name	Conduct the welding process and weld the work piece
Sector	Mining
Sub-Sector	Engineering Services
Occupation	Mechanical Services
NSQF Level	4
Credits	4
Version	2.0
Last Reviewed Date	08/05/2025
Next Review Date	06/05/2028
NSQC Clearance Date	08/05/2025

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MIN/N3203: Post welding operations activities

Description

This OS unit is about conducting quality checks and inspection of the finished products produced and repair the bad quality items produced.

Scope

The scope covers the following :

- Inspection of finished goods to detect any deviations from the product design
- Maintain records for production and defective pieces
- Unload and store the finished goods
- Ensure cleanliness and 5S is maintained at the workplace
- Conduct regular preventive maintenance of equipment

Elements and Performance Criteria

Inspection of finished goods to detect any deviations from the product design

To be competent, the user/individual on the job must be able to:

- PC1.** inspect the output products by comparing the dimensions of the output pieces with the specifications of the finished product using various measuring devices like micrometers, vernier calipers, gauges, rulers, weighing scales and etc.
- PC2.** compare texture, color, surface properties, hardness and strength with the given specifications described the in work order/ work Instructions
- PC3.** separate the defective pieces into two categories e.i. repairable and beyond repair by putting tags/ markings on the welded jig/ workpiece surface
- PC4.** ensure that the pieces which are not meet the specified standard and cannot be repaired are discarded

Maintain records for production and defective pieces

To be competent, the user/individual on the job must be able to:

- PC5.** maintain data records for quality defects and pieces which are beyond repair
- PC6.** prepare all documentation correctly on time
- PC7.** report completion of job allocated during the shift, problems encountered and further actions that need to be taken

Unload and store the finished goods

To be competent, the user/individual on the job must be able to:

- PC8.** ensure that the output pieces is correctly clamped and lifted using suitable equipment like hoist, lifts, crane, etc.
- PC9.** ensure that there is no damage to the lifted work pieces
- PC10.** carry the output product to the designated area using hangars, conveyor belts, cranes, forklifts etc.

Ensure cleanliness and 5S is maintained at the workplace

To be competent, the user/individual on the job must be able to:

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- PC11.** store all equipment in a proper order as indicated in the equipment manual and the designated area
- PC12.** ensure that the equipment and the work place are regularly cleaned and that there is not accumulation of dust, moisture and waste material
- PC13.** follow "5-S" practice at work place

Conduct regular preventive maintenance of equipment

To be competent, the user/individual on the job must be able to:

- PC14.** check the working of all bearing, rollers, shafts etc. and oil all moving parts of the equipment on a periodic basis
- PC15.** check the working of non-moving parts and periodically conduct preventive maintenance to prevent machine failure
- PC16.** check the equipment calibration periodically and report any errors to the maintenance teams for rectification

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** job-specific documents e.g. daily maintenance checklist and importance of the same
- KU2.** risk and impact of not following defined procedures/work instructions
- KU3.** the hierarchy for reporting identified problems
- KU4.** cost of equipment and loss for the company that results from damage of equipment
- KU5.** implications of delays in the process
- KU6.** handover and takeover procedures of the mine welder according to company's SOP
- KU7.** safety guidelines specified by Directorate General of Mines Safety (DGMS) specific to mine welding operations
- KU8.** different types of mines and detail of the mine one is working in
- KU9.** benching in quarries, dressing of overhangs, undercuts, fencing
- KU10.** importance of first aid and hygiene
- KU11.** code of practice in specific areas of the mine
- KU12.** standing orders in force at the mine
- KU13.** importance of safety in the vicinity of machinery
- KU14.** about shot-firing / blasting related safety regulations including taking shelter during blasting
- KU15.** duties of workmen under the Mines Act-1952
- KU16.** provision of compensation and working hours, leaves, etc. as per Mines Act-1952
- KU17.** the outcome of violation of safety procedures
- KU18.** emergency response /disaster management plan prepared by the organization
- KU19.** techniques of using measurement instruments like rulers, Vernier callipers, micrometres, weighing scale, gauges and other inspection equipment
- KU20.** guidelines to identify quality defects in work pieces visual/ test based
- KU21.** methods used for cutting, shearing, hammering, drilling which can repair pieces with minor defects

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- KU22.** basic level maintenance and cleaning techniques
- KU23.** various solvents, chemicals, lubricants etc used during the maintenance processes
- KU24.** procedure for arranging the equipment in the prescribed manner including tagging and numbering of machine parts
- KU25.** safety precautions to be taken during cleaning and maintenance activities
- KU26.** basic welding defects and corrective measures
- KU27.** basic level operations of lifting equipment like hoists, cranes, pulley etc.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** note down observations (if any)
- GS2.** read and interpret symbols and readings
- GS3.** read information documents
- GS4.** discuss task lists, schedules and activities
- GS5.** effectively communicate, listen and comprehend the information given by various sources about the site
- GS6.** make decisions pertaining to the concerned area of work
- GS7.** plan and organize the work order and tasks
- GS8.** organize all operation and service manuals so that sorting/ accessing information is easy
- GS9.** detect problems in day to day tasks
- GS10.** discuss possible solution with the supervisor for problem solving
- GS11.** make decisions in emergency conditions
- GS12.** follow instructions and work on areas of improvement identified
- GS13.** complete the assigned tasks timely
- GS14.** use reasoning skills to identify and resolve basic problems
- GS15.** analyze and detect any potential problems which could arise during operation
- GS16.** respect persons with disabilities
- GS17.** explain the importance of gender-sensitization at work site
- GS18.** state basic laws, acts and provisions defined for Persons with Disability (PwD) by the statutory bodies

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Inspection of finished goods to detect any deviations from the product design</i>	8	12	-	5
PC1. inspect the output products by comparing the dimensions of the output pieces with the specifications of the finished product using various measuring devices like micrometers, vernier calipers, gauges, rulers, weighing scales and etc.	2	3	-	2
PC2. compare texture, color, surface properties, hardness and strength with the given specifications described the in work order/ work Instructions	2	3	-	1
PC3. separate the defective pieces into two categories e.i. repairable and beyond repair by putting tags/ markings on the welded jig/ workpiece surface	2	3	-	1
PC4. ensure that the pieces which are not meet the specified standard and cannot be repaired are discarded	2	3	-	1
<i>Maintain records for production and defective pieces</i>	4	9	-	3
PC5. maintain data records for quality defects and pieces which are beyond repair	2	3	-	1
PC6. prepare all documentation correctly on time	1	3	-	1
PC7. report completion of job allocated during the shift, problems encountered and further actions that need to be taken	1	3	-	1
<i>Unload and store the finished goods</i>	6	11	-	5
PC8. ensure that the output pieces is correctly clamped and lifted using suitable equipment like hoist, lifts, crane, etc.	2	4	-	2
PC9. ensure that there is no damage to the lifted work pieces	2	4	-	2

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. carry the output product to the designated area using hangars, conveyor belts, cranes, forklifts etc.	2	3	-	1
<i>Ensure cleanliness and 5S is maintained at the workplace</i>	6	9	-	3
PC11. store all equipment in a proper order as indicated in the equipment manual and the designated area	2	3	-	1
PC12. ensure that the equipment and the work place are regularly cleaned and that there is not accumulation of dust, moisture and waste material	2	3	-	1
PC13. follow "5-S" practice at work place	2	3	-	1
<i>Conduct regular preventive maintenance of equipment</i>	6	9	-	4
PC14. check the working of all bearing, rollers, shafts etc. and oil all moving parts of the equipment on a periodic basis	2	3	-	2
PC15. check the working of non-moving parts and periodically conduct preventive maintenance to prevent machine failure	2	3	-	1
PC16. check the equipment calibration periodically and report any errors to the maintenance teams for rectification	2	3	-	1
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N3203
NOS Name	Post welding operations activities
Sector	Mining
Sub-Sector	Engineering Services
Occupation	Mechanical Services
NSQF Level	4
Credits	4
Version	2.0
Last Reviewed Date	08/05/2025
Next Review Date	06/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

Qualification Pack

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

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- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Qualification Pack

MIN/N1702: Follow Health, Safety and Environmental guidelines for Underground Metalliferous Mines

Description

This unit is about adhering to health, safety and environmental guidelines in Underground Metalliferous Mines (UMM) during the execution of various tasks and operations and maintenance.

Scope

The scope covers the following :

- Follow work-site health and safety measures
- Follow environmental guidelines

Elements and Performance Criteria

Follow work-site health and safety measures

To be competent, the user/individual on the job must be able to:

- PC1.** undertake "The Take-5 (Personal Risk Assessment)" before commencement of any work
- PC2.** comply with safety, health and security-related regulations/guidelines at the mine e.g. follow Standard Operating Procedure (SOP) for material handling in underground (U/G) mine
- PC3.** operate various types of fire extinguishers to control different types of fire at a worksite when required
- PC4.** check that roof supporting is as per Systematic Support Plan (SSP) and approved Systematic Support Rules (SSR) while undertaking work in an area
- PC5.** take precaution against occupational health hazards (like dust, water, mine gases etc.) due to U/G working environment
- PC6.** use self-rescue apparatus, appropriately when required
- PC7.** follow appropriate emergency response procedure during emergency such as fire, water inrush, fall of ground etc.
- PC8.** follow precautions against U/G electrical appliances
- PC9.** follow appropriate Standard Operating Procedure while working near any isolated and sealed off area of the mine
- PC10.** follow appropriate safety practices while traveling on U/G haul roads, incase of post blast fumes and misfire
- PC11.** follow the manufacturer's instructions for care and safe operation of mine machinery and equipment
- PC12.** identify the working ventilation district from line diagram of ventilation circuit to direct air to the working face
- PC13.** follow Safety Management Plan (SMP) and Emergency Management Plan (EMP)
- PC14.** follow gas detecting alarm signal on leakage of inflammable gases and laid out procedure to ensure safety
- PC15.** follow process for reporting any unsafe act/condition in work area to the concerned person

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- PC16.** use underground mine communication system
- PC17.** ensure positive isolation near the work place if applicable
- PC18.** use appropriate PPE as per the requirement
- PC19.** maintain hand hygiene by washing hands with alcohol based sanitisers/soap
- PC20.** disinfect the machine/tools before and after work/task
- PC21.** maintain hygiene at the work site
- PC22.** report any symptoms of illness to the shift-incharge
- PC23.** identify six directional hazards at workplace and take decisions accordingly

Follow environmental guidelines

To be competent, the user/individual on the job must be able to:

- PC24.** identify the environmental impact of mining related operations and follow steps to reduce those impact
- PC25.** follow the mineral conservation practices in U/G mining operations to achieve optimum ore or mineral recovery
- PC26.** ensure that the stowing practices produce minimum disturbance to the surface
- PC27.** ensure that the subgrade ore is carried out to surface and stacked separately at the earmarked place
- PC28.** ensure the productivity of the machine for material/fuel conservation
- PC29.** follow the process for collecting, storing and disposing of the hazardous material and waste (like used oil, lubricant, battery, etc.) in compliance with worksite guidelines
- PC30.** follow the "5-S" practice at work site like cleaning oil from ground (to avoid soil from getting damaged), etc.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** safety guidelines specified by Directorate General of Mine Safety (DGMS)
- KU2.** duties and rights of workers
- KU3.** selection process of person for rescue training
- KU4.** about Systematic Support Plan (SSP) and approved Systematic Support Rules (SSR)
- KU5.** isolation and sealed off area of the mine
- KU6.** various types of gases found in the mine and their effect
- KU7.** self-rescue apparatus and their uses
- KU8.** provision of medical examination (IME & PME) of person employed as per Mines Rules 1955
- KU9.** importance of first aid and hygiene
- KU10.** importance of blind spot survey/training
- KU11.** about different types of machinery used in U/G mines
- KU12.** different types of supporting systems used in U/G mines as per SSP and SSR
- KU13.** about precautions to be taken when handling heavy equipment
- KU14.** various problems/incidents likely to occur
- KU15.** role of Internal Safety Organization, safety committee, workman's inspector and DGMS

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- KU16.** about mine safety standard including light illumination level, noise levels, dust level, pollutants, etc at the work-site
- KU17.** common sources of pollution in the mines and ways to minimize it
- KU18.** various types of fire extinguishers
- KU19.** safety equipment like safety shoes, safety belt, tight fit clothing, hand gloves, safety goggles, Gas Detector, Safety Lamp, Self-Contained Breathing Apparatus, gum boots, ear plugs, Face Mask, etc.
- KU20.** shot-firing / blasting related safety regulations including taking shelter during blasting
- KU21.** emergency response /disaster management plan prepared by the organization as per DGMS guideline
- KU22.** mining area-specific signs, and other safety and emergency signals
- KU23.** the outcome of violation of safety procedures
- KU24.** Take-5 (Personal Risk Assessment) training (DGMS Tech. circulars 2/2014)
- KU25.** rules and regulations for safety and security while handling hazardous materials
- KU26.** safety appliances and rescue equipment
- KU27.** importance and use of various communication system used in UG mines
- KU28.** importance of positive isolation at working site
- KU29.** safety and occupational health policy of organisation
- KU30.** six directional hazard identification process
- KU31.** basic personal and workplace hygiene
- KU32.** importance of FAB (Fresh Air Base)
- KU33.** basic provisions in Mines Creche Rules, 1966 (MCR) for females employed in the mines
- KU34.** the role and responsibilities of rescue room and rescue station and how to contact them in case of emergency
- KU35.** importance of taking shelter at the miner's station during blasting operation
- KU36.** importance of sensitization towards different genders and persons with disabilities (PWD)
- KU37.** importance of following infection control policies, '5-S' practices, and waste management
- KU38.** importance of water/material/energy conservation and management
- KU39.** importance of SMP and EMP, prepared by the organization, as per DGMS guideline

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** fill up documentation applicable to one's role
- GS2.** make decisions on the concerned area of work
- GS3.** read basic English language
- GS4.** read and interpret manuals, health, and safety instructions, memos, etc.
- GS5.** use the digital information from machine
- GS6.** use basic applications of a computer
- GS7.** plan and organize the work order and tasks
- GS8.** use reasoning skills to identify and resolve fundamental problems

Qualification Pack

GS9. complete the assigned tasks timely

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Follow work-site health and safety measures</i>	24	42	-	17
PC1. undertake "The Take-5 (Personal Risk Assessment)" before commencement of any work	-	2	-	2
PC2. comply with safety, health and security-related regulations/guidelines at the mine e.g. follow Standard Operating Procedure (SOP) for material handling in underground (U/G) mine	1	3	-	-
PC3. operate various types of fire extinguishers to control different types of fire at a worksite when required	1	4	-	1
PC4. check that roof supporting is as per Systematic Support Plan (SSP) and approved Systematic Support Rules (SSR) while undertaking work in an area	1	1	-	-
PC5. take precaution against occupational health hazards (like dust, water, mine gases etc.) due to U/G working environment	1	2	-	1
PC6. use self-rescue apparatus, appropriately when required	-	4	-	1
PC7. follow appropriate emergency response procedure during emergency such as fire, water inrush, fall of ground etc.	1	2	-	1
PC8. follow precautions against U/G electrical appliances	2	2	-	2
PC9. follow appropriate Standard Operating Procedure while working near any isolated and sealed off area of the mine	2	2	-	1
PC10. follow appropriate safety practices while traveling on U/G haul roads, incase of post blast fumes and misfire	2	4	-	-
PC11. follow the manufacturer's instructions for care and safe operation of mine machinery and equipment	1	2	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. identify the working ventilation district from line diagram of ventilation circuit to direct air to the working face	1	1	-	-
PC13. follow Safety Management Plan (SMP) and Emergency Management Plan (EMP)	1	-	-	-
PC14. follow gas detecting alarm signal on leakage of inflammable gases and laid out procedure to ensure safety	1	2	-	1
PC15. follow process for reporting any unsafe act/condition in work area to the concerned person	-	2	-	1
PC16. use underground mine communication system	-	1	-	-
PC17. ensure positive isolation near the work place if applicable	1	1	-	1
PC18. use appropriate PPE as per the requirement	2	4	-	2
PC19. maintain hand hygiene by washing hands with alcohol based sanitisers/soap	1	1	-	1
PC20. disinfect the machine/tools before and after work/task	1	1	-	-
PC21. maintain hygiene at the work site	1	1	-	1
PC22. report any symptoms of illness to the shift-incharge	1	-	-	-
PC23. identify six directional hazards at workplace and take decisions accordingly	2	-	-	-
<i>Follow environmental guidelines</i>	6	8	-	3
PC24. identify the environmental impact of mining related operations and follow steps to reduce those impact	1	2	-	-
PC25. follow the mineral conservation practices in U/G mining operations to achieve optimum ore or mineral recovery	-	1	-	-
PC26. ensure that the stowing practices produce minimum disturbance to the surface	1	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC27. ensure that the subgrade ore is carried out to surface and stacked separately at the earmarked place	1	-	-	1
PC28. ensure the productivity of the machine for material/fuel conservation	1	1	-	-
PC29. follow the process for collecting, storing and disposing of the hazardous material and waste (like used oil, lubricant, battery, etc.) in compliance with worksite guidelines	1	1	-	1
PC30. follow the "5-S" practice at work site like cleaning oil from ground (to avoid soil from getting damaged), etc.	1	2	-	1
NOS Total	30	50	-	20



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1702
NOS Name	Follow Health, Safety and Environmental guidelines for Underground Metalliferous Mines
Sector	Mining
Sub-Sector	Mining Operation
Occupation	HSE Functions, Mine Surveying, Drilling/Cutting, Shot firing/Blasting, Loading and Hauling - Underground, Specialized Operations, Electrical Services, Mechanical Services
NSQF Level	3.5
Credits	2
Version	2.0
Last Reviewed Date	25/02/2026
Next Review Date	25/02/2029
NSQF Clearance Date	25/02/2026

Qualification Pack

MIN/N1703: Follow Health, Safety, and Environmental Guidelines for opencast mines

Description

This unit is about adhering to health, safety and environmental guidelines at the Open Cast Mines while executing various tasks, maintenance and operations.

Scope

The scope covers the following :

- Follow work-site health and safety measures
- Follow Environmental guidelines

Elements and Performance Criteria

Follow work-site health and safety measures

To be competent, the user/individual on the job must be able to:

- PC1.** comply with safety, health, and security-related regulations/guidelines at the opencast mine
- PC2.** follow the safety instructions given by the workman's inspector
- PC3.** follow adequate safety while working at haul roads, heights, overburden dumps, sump area, stockyard, near moving parts, etc.
- PC4.** take safety precautions while working on sites (sub-station, workshop etc.), with equipment, and conducting welding and cutting operations
- PC5.** follow appropriate Safe Operating Procedure (SOP) while dealing with explosives
- PC6.** respond promptly and appropriately to an accident/ incident or an emergency situation, within limits of the role and responsibility
- PC7.** provide first aid to an injured person
- PC8.** operate various types of fire extinguishers to control different types of fire at a worksite when required
- PC9.** use appropriate PPE as per the requirement
- PC10.** maintain hand hygiene by washing hands with alcohol based sanitisers/soap
- PC11.** disinfect the machine/tools before and after work/task
- PC12.** maintain hygiene at the work site
- PC13.** report any symptoms of illness to the shift-incharge

Follow Environmental guidelines

To be competent, the user/individual on the job must be able to:

- PC14.** identify the environmental impact of related opencast mining operations
- PC15.** follow the process for collecting, storing and disposing of the hazardous material and waste (like used oil, lubricant, battery, etc.) in compliance with worksite guidelines
- PC16.** ensure not to mix topsoil with waste in day to day tasks
- PC17.** ensure that HEMM is washed at the designated location

Qualification Pack

- PC18.** ensure the productivity of the machine for material/fuel conservation
- PC19.** follow the mineral conservation practices specified by the organization in accordance with MCDR-2017 (Mineral Conservation and Development Rules)
- PC20.** assist supervisor for reducing environmental impact caused due to related mining operations

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** about various environmental awareness program related to mining, organized by the various government bodies/company
- KU2.** safety guidelines specified by Directorate General of Mine Safety (DGMS)
- KU3.** how to identify six-directional hazards
- KU4.** basic mining terminologies and definitions
- KU5.** importance of blind spot survey/training
- KU6.** means of access and egress from the mines, location of workshop, haul roads and working face including dump yards
- KU7.** duties of workers under The Mines act-1952
- KU8.** working hours and accident compensation as per The Mines act-1952
- KU9.** about precautions to be taken when handling heavy equipment
- KU10.** various problems/incidents likely to occur
- KU11.** hierarchy of the reporting
- KU12.** machine operation, condition of the machine and worksite
- KU13.** proper documents specific to the machine
- KU14.** role of workmen inspector, safety committee and internal safety organization
- KU15.** the process of top soil removal and management
- KU16.** mine sump and pumping system of the mines
- KU17.** about mine safety standard including light illumination level, noise levels, dust level, pollutants, etc at the work-site
- KU18.** common sources of pollution in the mines and ways to minimize it
- KU19.** safety equipment like safety shoes, safety belt, tight fit clothing, hand gloves, safety goggles, gas detector, safety lamp, self-contained breathing apparatus, gum boots, ear plugs, face mask, etc.
- KU20.** shot-firing / blasting related safety regulations including taking shelter during blasting
- KU21.** emergency response /disaster management plan prepared by the organization
- KU22.** signages, mining area-specific signs, and other safety and emergency signals
- KU23.** the outcome of violation of safety procedures
- KU24.** basic personal and workplace hygiene
- KU25.** importance of sensitization towards different genders and PWD (Persons with Disabilities)

Generic Skills (GS)

User/individual on the job needs to know how to:



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- GS1.** fill up documentation applicable to one's role
- GS2.** make decisions on the concerned area of work
- GS3.** read and interpret manuals, health, and safety instructions, memos, etc.
- GS4.** plan and organize the work order and jobs
- GS5.** use reasoning skills to identify and resolve fundamental problems
- GS6.** complete the assigned tasks timely

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Follow work-site health and safety measures</i>	20	34	-	15
PC1. comply with safety, health, and security-related regulations/guidelines at the opencast mine	2	2	-	1
PC2. follow the safety instructions given by the workman's inspector	1	3	-	1
PC3. follow adequate safety while working at haul roads, heights, overburden dumps, sump area, stockyard, near moving parts, etc.	3	3	-	1
PC4. take safety precautions while working on sites (sub-station, workshop etc.), with equipment, and conducting welding and cutting operations	1	3	-	1
PC5. follow appropriate Safe Operating Procedure (SOP) while dealing with explosives	2	3	-	2
PC6. respond promptly and appropriately to an accident/ incident or an emergency situation, within limits of the role and responsibility	2	3	-	1
PC7. provide first aid to an injured person	2	3	-	1
PC8. operate various types of fire extinguishers to control different types of fire at a worksite when required	1	3	-	1
PC9. use appropriate PPE as per the requirement	2	4	-	2
PC10. maintain hand hygiene by washing hands with alcohol based sanitisers/soap	1	2	-	1
PC11. disinfect the machine/tools before and after work/task	1	2	-	1
PC12. maintain hygiene at the work site	1	1	-	1
PC13. report any symptoms of illness to the shift-incharge	1	2	-	1
<i>Follow Environmental guidelines</i>	10	16	-	5

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. identify the environmental impact of related opencast mining operations	2	2	-	1
PC15. follow the process for collecting, storing and disposing of the hazardous material and waste (like used oil, lubricant, battery, etc.) in compliance with worksite guidelines	1	2	-	1
PC16. ensure not to mix topsoil with waste in day to day tasks	2	2	-	1
PC17. ensure that HEMM is washed at the designated location	2	2	-	1
PC18. ensure the productivity of the machine for material/fuel conservation	1	3	-	-
PC19. follow the mineral conservation practices specified by the organization in accordance with MCDR-2017 (Mineral Conservation and Development Rules)	1	3	-	1
PC20. assist supervisor for reducing environmental impact caused due to related mining operations	1	2	-	-
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1703
NOS Name	Follow Health, Safety, and Environmental Guidelines for opencast mines
Sector	Mining
Sub-Sector	Mining Operation
Occupation	HSE Functions, , Exploration, Mineral Estimation, Planning, Mine Surveying, Drilling/Cutting, Shot firing/Blasting, Loading and Hauling - Opencast, Specialized Operations, Electrical Services, Mechanical Services, Instrumentation and Control Systems, Ore Processing
NSQF Level	3.5
Credits	2
Version	2.0
Last Reviewed Date	11/03/2026
Next Review Date	11/03/2029
NSQF Clearance Date	11/03/2026

Qualification Pack

MIN/N1704: Follow Health, Safety, and Environmental guidelines for underground coal mines

Description

This unit is about adhering to health, safety and environmental guidelines in Underground Coal Mines (UCM) during execution of various tasks, operations and maintenance.

Scope

The scope covers the following :

- Follow work-site health and safety measures
- Follow environmental guidelines

Elements and Performance Criteria

Follow work-site health and safety measures

To be competent, the user/individual on the job must be able to:

- PC1.** follow preventive measures against firedamp, whitedamp, blackdamp etc.
- PC2.** use the flame safety lamp for detecting the methane gas as per Standard Operating Procedure (SOP)
- PC3.** undertake "The Take-5 (Personal Risk Assessment)" before commencement of any work
- PC4.** comply with safety, health and security-related regulations/guidelines at the mine e.g. SOP for material handling in underground (U/G) mine
- PC5.** ensure that oil, grease, canvas or other inflammable material are stored in fire-proof receptacle
- PC6.** ensure that every instrument, apparatus and equipment are DGMS approved before these are used
- PC7.** ensure that Armoured face conveyor (AFC) and chocks must be kept in a straight line for every cycle of operations and tightened up to the setting pressure while keeping it in full contact with the roof, applicable for longwall mining
- PC8.** provide first aid to an injured person
- PC9.** follow safety precautions against spontaneous heating of the coal
- PC10.** operate various types of fire extinguishers to control different types of fire at worksite, if required
- PC11.** ensure that no person is traveling/working/staying under unsupported roof
- PC12.** check that roof supporting is as per Systematic Support Plan (SSP) and approved Systematic Support Rules (SSR) while undertaking work in an area
- PC13.** take precaution against occupational health hazards (like dust, water, mine gases etc.) due to U/G working environment
- PC14.** use self-rescue apparatus appropriately when required
- PC15.** follow Safety Management Plan (SMP) and Emergency Management Plan (EMP)
- PC16.** follow precautions against U/G electrical appliances

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- PC17.** take proper care against damage and accidents while loading, transporting, dismantling and erecting of roof supports
- PC18.** follow appropriate SOP while working near any isolated and sealed off area of the mine
- PC19.** ensure that the roof and the sidewalls of the mine face (or newly exposed area of the mines) have been scaled/dressed properly
- PC20.** take relevant safety precautions during depillaring operation in UCM
- PC21.** follow appropriate safety practices while traveling on U/G haul roads, incase of post blast fumes and misfire
- PC22.** follow the manufacturer's instructions for care and safe operation of mine machinery and equipment
- PC23.** identify the working ventilation district from line diagram of ventilation circuit to direct air to the working face
- PC24.** follow laid out SOP in case of alarm signal for leakage of inflammable gases
- PC25.** follow the process of reporting any unsafe act/condition in the working area to the concerned person
- PC26.** use underground mine communication system
- PC27.** ensure positive isolation near the work place if applicable
- PC28.** use appropriate Personal Protective Equipment (PPE) as per the requirement
- PC29.** maintain hand hygiene by washing hands with alcohol based sanitisers/soap
- PC30.** disinfect the machine/tools before and after work/task
- PC31.** maintain hygiene at the work site
- PC32.** report any symptoms of illness to the shift-incharge
- PC33.** identify six directional hazards at workplace and take decisions accordingly

Follow environmental guidelines

To be competent, the user/individual on the job must be able to:

- PC34.** identify the environmental impact of mining related operations and follow steps to reduce those impact
- PC35.** follow the mineral conservation practices in U/G mining operations to achieve optimum ore or mineral recovery
- PC36.** ensure that the stowing practices produce minimum disturbance to the surface
- PC37.** ensure that the subgrade coal is carried out to surface and stacked separately at the earmarked place
- PC38.** ensure the productivity of the machine for material/fuel conservation
- PC39.** follow the process for collecting, storing and disposing of the hazardous material and waste (like used oil, lubricant, battery, etc.) in compliance with worksite guidelines
- PC40.** follow the "5-S" practice at work site like cleaning oil from ground (to avoid soil from getting damaged), etc.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** safety guidelines specified by Directorate General of Mine Safety (DGMS)

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- KU2.** duties and rights of workers
- KU3.** selection process of person for rescue training
- KU4.** about Systematic Support Plan (SSP) and approved Systematic Support Rules (SSR)
- KU5.** isolation and sealed off area of the mine
- KU6.** various types of gases available in the mine and their effects; and their control measures
- KU7.** self-rescue apparatus and their uses
- KU8.** provision of medical examination (Initial Medical Examination (IME) & Periodical Medical Examination (PME)) of a person employed, as per Mines Rules 1955
- KU9.** importance of first aid and hygiene
- KU10.** about different types of machineries used in U/G mines
- KU11.** different types of supporting system used in U/G mines as per SSP and SSR
- KU12.** about precautions to be taken when handling heavy equipment
- KU13.** various problems/incidents likely to occur
- KU14.** role of Internal Safety Organization, safety committee, workman's inspector and DGMS
- KU15.** about mine safety standard including light illumination level, noise levels, dust level, pollutants, etc at the work-site
- KU16.** common sources of pollution in the mines and ways to minimize it
- KU17.** various types of fire extinguishers
- KU18.** safety equipment like safety shoes, safety belt, tight fit clothing, hand gloves, safety goggles, Gas Detector, Safety Lamp, Self-Contained Breathing Apparatus, gum boots, ear plugs, Face Mask, etc.
- KU19.** shot-firing / blasting related safety regulations including taking shelter during blasting
- KU20.** mining area-specific signs, and other safety and emergency signals
- KU21.** the outcome of violation of safety procedures
- KU22.** Take-5 (Personal Risk Assessment) training (DGMS Tech. circulars 2/2014)
- KU23.** hazardous material safety, security rules and regulations
- KU24.** safety appliances and rescue equipment
- KU25.** importance and use of various communication system used in UG mines
- KU26.** importance of positive isolation at working site
- KU27.** safety and occupational health policy of organisation
- KU28.** six directional hazard identification process
- KU29.** importance of blind spot survey/training
- KU30.** basic personal and workplace hygiene
- KU31.** importance of FAB (Fresh Air Base)
- KU32.** basic provisions in Mines Creche Rules, 1966 (MCR) for any females employed in the mines
- KU33.** about basic safety regulations of Coal Mines Regulation, 2017 (CMR)
- KU34.** types of stone dust barrier and its importance
- KU35.** coal dust explosion and its preventive measures
- KU36.** classification of coal mines as per the degree of gassiness of coal seams such as first degree, second degree, and third-degree mines
- KU37.** precautions as per the gassiness of the coal mines

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- KU38.** use of flame safety lamp and its parts
- KU39.** about coal mines occupational disease such as pneumoconiosis or 'black lung' and their preventive measures
- KU40.** Standard of Ventilation as per the section 153 of the CMR 2017
- KU41.** Standard of Lighting as per the section 178 of the CMR 2017
- KU42.** the roles, duties and responsibilities of rescue team members, rescue room and rescue station and how to contact them in case of emergency
- KU43.** the correct steps for conducting any rescue work as per Mine Rescue Rule (MRR)
- KU44.** importance of taking shelter at the miner's station during blasting operation
- KU45.** importance of sensitization towards different genders and persons with disabilities (PWD)
- KU46.** importance of following infection control policies, '5-S' practices, and waste management
- KU47.** importance of water/material/energy conservation and management
- KU48.** importance of SMP and EMP, prepared by the organization, as per DGMS guideline

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** fill up documentation applicable to one's role
- GS2.** make decisions on the concerned area of work
- GS3.** read basic English language
- GS4.** read and interpret manuals, health, and safety instructions, memos, etc.
- GS5.** use the digital information from a machine
- GS6.** plan and organize the work order and tasks
- GS7.** use basic applications of the computer
- GS8.** use reasoning skills to identify and resolve fundamental problems
- GS9.** complete the assigned tasks timely

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Follow work-site health and safety measures</i>	24	43	-	14
PC1. follow preventive measures against firedamp, whitedamp, blackdamp etc.	1	1	-	-
PC2. use the flame safety lamp for detecting the methane gas as per Standard Operating Procedure (SOP)	-	1	-	-
PC3. undertake "The Take-5 (Personal Risk Assessment)" before commencement of any work	1	1	-	1
PC4. comply with safety, health and security-related regulations/guidelines at the mine e.g. SOP for material handling in underground (U/G) mine	1	1	-	1
PC5. ensure that oil, grease, canvas or other inflammable material are stored in fire-proof receptacle	-	1	-	-
PC6. ensure that every instrument, apparatus and equipment are DGMS approved before these are used	1	1	-	-
PC7. ensure that Armoured face conveyor (AFC) and chocks must be kept in a straight line for every cycle of operations and tightened up to the setting pressure while keeping it in full contact with the roof, applicable for longwall mining	3	3	-	-
PC8. provide first aid to an injured person	1	2	-	1
PC9. follow safety precautions against spontaneous heating of the coal	1	1	-	-
PC10. operate various types of fire extinguishers to control different types of fire at worksite, if required	-	2	-	-
PC11. ensure that no person is traveling/working/staying under unsupported roof	1	-	-	-
PC12. check that roof supporting is as per Systematic Support Plan (SSP) and approved Systematic Support Rules (SSR) while undertaking work in an area	-	2	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. take precaution against occupational health hazards (like dust, water, mine gases etc.) due to U/G working environment	-	1	-	1
PC14. use self-rescue apparatus appropriately when required	-	2	-	-
PC15. follow Safety Management Plan (SMP) and Emergency Management Plan (EMP)	1	2	-	1
PC16. follow precautions against U/G electrical appliances	2	2	-	1
PC17. take proper care against damage and accidents while loading, transporting, dismantling and erecting of roof supports	1	2	-	-
PC18. follow appropriate SOP while working near any isolated and sealed off area of the mine	1	1	-	1
PC19. ensure that the roof and the sidewalls of the mine face (or newly exposed area of the mines) have been scaled/dressed properly	1	1	-	-
PC20. take relevant safety precautions during depillaring operation in UCM	-	2	-	1
PC21. follow appropriate safety practices while traveling on U/G haul roads, incase of post blast fumes and misfire	1	1	-	-
PC22. follow the manufacturer's instructions for care and safe operation of mine machinery and equipment	-	1	-	1
PC23. identify the working ventilation district from line diagram of ventilation circuit to direct air to the working face	-	1	-	1
PC24. follow laid out SOP in case of alarm signal for leakage of inflammable gases	1	1	-	-
PC25. follow the process of reporting any unsafe act/condition in the working area to the concerned person	-	1	-	1
PC26. use underground mine communication system	-	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC27. ensure positive isolation near the work place if applicable	1	1	-	-
PC28. use appropriate Personal Protective Equipment (PPE) as per the requirement	2	2	-	1
PC29. maintain hand hygiene by washing hands with alcohol based sanitisers/soap	-	1	-	-
PC30. disinfect the machine/tools before and after work/task	1	1	-	-
PC31. maintain hygiene at the work site	1	1	-	-
PC32. report any symptoms of illness to the shift-incharge	1	1	-	-
PC33. identify six directional hazards at workplace and take decisions accordingly	-	1	-	1
<i>Follow environmental guidelines</i>	6	7	-	6
PC34. identify the environmental impact of mining related operations and follow steps to reduce those impact	1	1	-	1
PC35. follow the mineral conservation practices in U/G mining operations to achieve optimum ore or mineral recovery	1	1	-	1
PC36. ensure that the stowing practices produce minimum disturbance to the surface	1	1	-	-
PC37. ensure that the subgrade coal is carried out to surface and stacked separately at the earmarked place	1	1	-	1
PC38. ensure the productivity of the machine for material/fuel conservation	1	1	-	1
PC39. follow the process for collecting, storing and disposing of the hazardous material and waste (like used oil, lubricant, battery, etc.) in compliance with worksite guidelines	-	1	-	1
PC40. follow the "5-S" practice at work site like cleaning oil from ground (to avoid soil from getting damaged), etc.	1	1	-	1



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1704
NOS Name	Follow Health, Safety, and Environmental guidelines for underground coal mines
Sector	Mining
Sub-Sector	Mining Operation
Occupation	HSE Functions, Mine Surveying, Drilling/Cutting, Shot firing/Blasting, Loading and Hauling - Underground, Specialized Operations, Electrical Services, Mechanical Services
NSQF Level	3.5
Credits	2
Version	2.0
Last Reviewed Date	25/02/2026
Next Review Date	25/02/2029
NSQC Clearance Date	25/02/2026

Qualification Pack

MIN/N3204: Special preparation for Welding at U/G Mines

Description

This OS unit is about Special preparation for Welding at U/G Gassy Mines.

Scope

The scope covers the following :

- Special preparation for first & second degree gassy mines
- Special preparation for Third degree gassy mines

Elements and Performance Criteria

Special preparation for first & second degree gassy mines

To be competent, the user/individual on the job must be able to:

- PC1.** permission taken in writing from manager before going welding works at U/G mines
- PC2.** carried out the welding work as per the SOP
- PC3.** pre-inspection of site relating to safety aspects/arrangement i.e. presence of methane, availability of firefighting equipment, site/plan preferences where welding is to be carried out.
- PC4.** precaution taken during welding and cutting operation at U/G mines
- PC5.** carry out post inspection of site relating to safety
- PC6.** ensure that the pieces which are not as per requirement and not meeting the specified standard and cannot be repaired are discarded

Special preparation for third degree gassy mines

To be competent, the user/individual on the job must be able to:

- PC7.** ensure prior permission from regional inspector of mines before conducting any welding works in U/G mines
- PC8.** perform welding and cutting with close supervision of authorized supervisor

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** job-specific documents e.g. daily maintenance checklist and importance of the same
- KU2.** risk and impact of not following defined procedures/work instructions
- KU3.** CMR and MMR regulation related to welding and cutting work
- KU4.** the hierarchy for reporting identified problems
- KU5.** cost of equipment and loss for the company that results from damage of equipment
- KU6.** implications of delays in the process
- KU7.** importance of first aid and hygiene
- KU8.** code of practice in specific areas of the mine
- KU9.** standing orders in force at the mine

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- KU10.** duties of workmen under the Mines Act-1952
- KU11.** provision of compensation and working hours, leaves, etc. as per Mines Act-1952
- KU12.** about shot-firing / blasting related safety regulations including taking shelter during blasting
- KU13.** techniques of using measurement instruments like rulers, Vernier callipers, micrometres, weighing scale, gauges and other inspection equipment
- KU14.** guidelines to identify quality defects in work pieces -visual/ test based
- KU15.** methods used for cutting, shearing, hammering, drilling which can repair pieces with minor defects
- KU16.** basic level maintenance and cleaning techniques
- KU17.** various solvents, chemicals, lubricants etc used during the maintenance processes
- KU18.** procedure for arranging the equipment in the prescribed manner including tagging and numbering of machine parts
- KU19.** safety precautions to be taken during cleaning and maintenance activities
- KU20.** basic welding defects and corrective measures
- KU21.** basic level operations of lifting equipment like hoists, cranes, pulley etc.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** note down observations (if any)
- GS2.** read and interpret symbols and readings
- GS3.** read information documents
- GS4.** discuss task lists, schedules and activities
- GS5.** effectively communicate, listen and comprehend the information given by various sources about the site
- GS6.** make decisions pertaining to the concerned area of work
- GS7.** plan and organize the work order and tasks
- GS8.** organize all operation and service manuals so that sorting/ accessing information is easy
- GS9.** detect problems in day to day tasks
- GS10.** discuss possible solution with the supervisor for problem solving
- GS11.** make decisions in emergency conditions
- GS12.** follow instructions and work on areas of improvement identified
- GS13.** complete the assigned tasks timely
- GS14.** explain the importance of gender-sensitization at work site

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Special preparation for first & second degree gassy mines</i>	24	38	-	16
PC1. permission taken in writing from manager before going welding works at U/G mines	4	6	-	2
PC2. carried out the welding work as per the SOP	4	6	-	2
PC3. pre-inspection of site relating to safety aspects/arrangement i.e. presence of methane, availability of firefighting equipment, site/plan preferences where welding is to be carried out.	4	7	-	3
PC4. precaution taken during welding and cutting operation at U/G mines	4	7	-	3
PC5. carry out post inspection of site relating to safety	4	6	-	3
PC6. ensure that the pieces which are not as per requirement and not meeting the specified standard and cannot be repaired are discarded	4	6	-	3
<i>Special preparation for third degree gassy mines</i>	6	12	-	4
PC7. ensure prior permission from regional inspector of mines before conducting any welding works in U/G mines	3	6	-	2
PC8. perform welding and cutting with close supervision of authorized supervisor	3	6	-	2
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N3204
NOS Name	Special preparation for Welding at U/G Mines
Sector	Mining
Sub-Sector	Engineering Services
Occupation	Mechanical Services
NSQF Level	4
Credits	1
Version	2.0
Last Reviewed Date	08/05/2025
Next Review Date	06/05/2028
NSQC Clearance Date	08/05/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack

Qualification Pack

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MIN/N3201.Prepare the welding machine, auxiliaries and work pieces for the welding process	30	50	0	20	100	20
MIN/N3202.Conduct the welding process and weld the work piece	30	50	0	20	100	30
MIN/N3203.Post welding operations activities	30	50	0	20	100	20
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	10
Total	110	180	-	60	350	80

Elective: 1 Underground Metal

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MIN/N1702.Follow Health, Safety and Environmental guidelines for Underground Metalliferous Mines	30	50	-	20	100	20
Total	30	50	-	20	100	20

Elective: 2 Opencast

Qualification Pack

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MIN/N1703.Follow Health, Safety, and Environmental Guidelines for opencast mines	30	50	-	20	100	20
Total	30	50	-	20	100	20

Elective: 3 Underground Coal

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MIN/N1704.Follow Health, Safety, and Environmental guidelines for underground coal mines	30	50	-	20	100	20
Total	30	50	-	20	100	20

Optional: 1 Underground Gassy Mines

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MIN/N3204.Special preparation for Welding at U/G Mines	30	50	0	20	100	20
Total	30	50	-	20	100	20

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
NOS	National Occupational Standard(s)
QP	Qualification Pack
NSQF	National Skills Qualifications Framework
TVET	Technical and Vocational Education and Training
SCMS	Skill Council for Mining Sector
ROPS	Roll-Over Protection Structure
FOPS	Falling Object Protective Structure
PC	Performance Criteria
HEMM	Heavy Earth Moving Machinery
OEM	Original Equipment Manufacturer
DGMS	Directorate General of Mining Safety
MMR	Metalliferous Mines Regulations
CMR	Coal Mines Regulations
SOP	Standard Operating Practice
MVTR	Mine Vocational Training Rules
MRR	Mines Rescue Rules
HSE	health, safety & environment
OB	Over Burden
ERP	Enterprise resource planning
MCDR	Mineral Conservation and Development Rules

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.
Sector	A sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	a sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in the industry.
Occupational Standards	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standard	NOS are occupational standards which apply uniquely in the Indian context.

Qualification Pack

Qualification Pack	QP comprises the set of OS, together with the educational, training, and other criteria required to perform a job role. A QP is assigned a unique qualifications pack role
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.